UPCODE SOFTWARE LABS | DATA ANALYST CAPSTONE

GoldVista Retail Analytics

Project Report: eight business challenges across 18 branches, Jan 2025 – Feb 2026

Prepared by: Aliya Arakkal Ameer

Date: October 2026

Tools: PostgreSQL (goldvista_db), Excel, Power BI Desktop

Currency: Saudi Riyal (SAR).

1. Executive Summary

GoldVista sold **SAR 348.3 million** across **61,361 invoices** to **45,363 customers** between January 2025 and February 2026. This report answers eight business questions using PostgreSQL queries, four Excel workbooks and a six-page Power BI dashboard built on a star-schema model with 39 DAX measures.

Measure	Value
Revenue (sales only, returns excluded)	SAR 348.3M
Invoices	61,361
Customers	45,363 (16% bought more than once)
Return rate	1.75% of invoices (1,071 returns)
Old-gold buy-back payout	about SAR 37.8M

What the data says

- **Revenue is concentrated.** Branches 16, 02 and 01 bring in about 30% of revenue, and the 12 largest branches about 93%.
- **Most customers do not come back, but the ones who do are worth far more.** 84% bought once. The 16% who return produce about 43% of revenue and come back after about 32 days.
- **Friday is the peak day.** It carries 25% of invoices and 21% of revenue, but with the lowest average invoice of the week.
- **Buy-back customers are worth more.** Their average invoice is about 12% higher and they are three times as likely to come back (45.7% against 14.6%), yet buy-back is used very unevenly across branches.
- **Cash still dominates.** 64% of money collected is cash and 32% is card. Loyalty-point redemption is almost unused.
- **The product range is narrow at the top.** Five item types make up 75% of revenue, and 22K and 24K gold together make up about 90%.

Priority actions

- Launch a follow-up offer around day 30 after a first purchase to lift the repeat rate.
- Roll out the buy-back approach of Branches 02 and 05 to large branches that use it little, such as Branches 13 and 19.
- Staff and promote for Thursday to Saturday, and give every salesperson an individual login so performance can be measured fairly.

2. Business Context, Data and Approach

2.1 The business and the data

The data sits in a PostgreSQL database, **goldvista_db**, with 19 tables: 8 fact tables (sales, returns, old-gold purchases, three payment tables and two daily summaries) and 11 dimension tables (branch, customer, date, item, karat, salesperson and others). Sales has about 90,800 invoice lines. A full column-level description is in the data dictionary (docs/data_dictionary.xlsx).

2.2 Approach

- **SQL:** 18 business queries (window functions, CTEs, ranking, rolling averages), 8 exploration queries and 8 reusable views, each with a header comment stating the business question.
- **Excel:** four workbooks for branch revenue, customer reporting, operations and product analysis.
- **Power BI:** a star-schema model, 39 DAX measures (see docs/dax_measures.xlsx), six dashboard pages, drill-through, bookmarks and a custom theme.

Revenue figures use sales lines only (is_return = FALSE), so returns are not counted as revenue. Returns are analysed separately in Section 5.

2.3 How to read this report

Each of the eight sections follows the same pattern: the business question, what the data shows, a chart, and a recommendation. Sections 11 and 12 collect all recommendations and the data limitations.

3. C-01 Branch Performance

Business question: *Which branches drive revenue, and how is performance changing over time?*

Source queries: branch_ranking, branch_rolling_avg, branch_performance

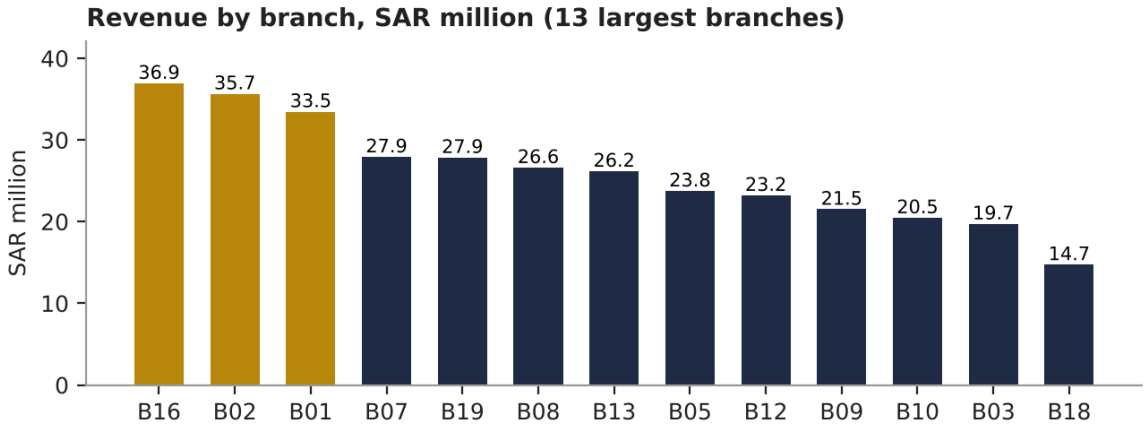


Figure 1. Revenue by branch (sales only). Gold bars mark the top three.

- **Branches 16 (SAR 36.9M), 02 (35.7M) and 01 (33.5M)** together produce about 30% of revenue. The top 12 branches produce about 93%.
- **Volume and value differ.** Branch 08 handles the most invoices (10,286) but has one of the lowest average invoice values (about SAR 2,650). Branch 02 has the highest average among the large branches (about SAR 5,480).
- **Weight is not revenue.** Branch 18 sells the most weight (about 736,000 g) but ranks 13th by revenue, because much of its volume is low-value silver.
- **Several branches have partial histories.** Branches 04, 06, 15 and 17 only have data for January to March 2025, Branch 03 starts in July 2025, Branch 20 starts in January 2026 (12 invoices) and Branches 11 and 14 have no sales. Comparing them with full-period branches is not like-for-like.
- **Branch 18 distorts growth.** Its monthly revenue jumped from SAR 0.2M in December 2025 to SAR 7.1M in January 2026, which is why a simple first-half versus second-half growth measure shows extreme values for some branches.

Possible causes: Branch size and opening dates explain part of the gap, since several branches have partial histories. Product mix also matters: Branch 18 sells bulk silver, and Branch 08 makes many low-value sales. Location and staffing were not analysed.

> **Recommendation:** Benchmark branches on **revenue per active month** and average invoice value, not on total revenue alone. Investigate why Branches 04, 06, 15 and 17 stop reporting after March 2025 (closure or missing data).

4. C-02 Customer Retention

Business question: *How many customers come back, how soon, and where is retention strongest?*

Source queries: customer_behavior, customer_behavior_by_branch, repeat_gap, buyback_repeat_rate, customer_value_by_segment

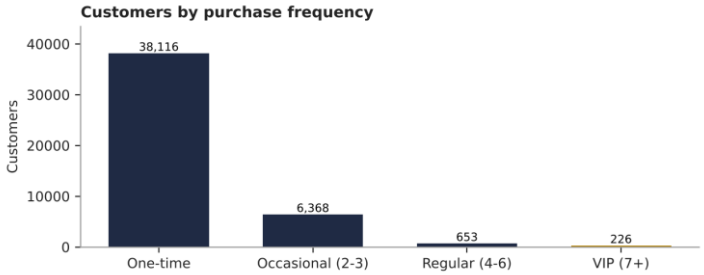


Figure 2. Customers by number of invoices.

- **84% of customers (38,116) bought once.** Another 6,368 (14%) bought two or three times, 653 (1.4%) four to six times and only 226 (0.5%) seven or more.
- **Repeat rates vary by branch.** Among the larger branches, Branch 02 (27.3%) and Branch 12 (23.0%) do best, while Branch 13 (8.8%) and Branch 16 (10.7%) do worst. Branch 16 is the top revenue branch, so its low repeat rate points to a retention opportunity. Branch 05 shows 30.3% but has only 109 customers and about 4,480 invoices, which suggests a few generic customer records rather than genuine loyalty.
- **Few customers return in any given month.** In the dashboard cohort view, about 2% of a cohort buys in the first month after its first purchase, about 1% in months 2 to 5, and almost none after month 6.
- **7,247 customers (16.0%) are repeat buyers and are worth far more.** They made their second purchase **31.7 days** after the first on average, and generate about SAR 149M, or 43% of revenue.

Segment	Customers	Revenue (SAR M)	Share of revenue	Avg revenue per customer (SAR)
One-time	38,116	198.8	57.1%	5,217
Occasional (2-3)	6,368	88.7	25.5%	13,935
Regular (4-6)	653	21.0	6.0%	32,101
VIP (7+)	226	39.8	11.4%	176,048

- **Buy-back customers return far more often:** 45.7% (917 of 2,007) bought again, against 14.6% (6,330 of 43,356) of other customers.

Value is revenue over 14 months, not true lifetime value. The VIP group and the buy-back rate may be inflated by generic customer records (the top customer account has 993 invoices worth SAR 5.8M, and Branch 05 has 109 customers for about 4,480 invoices).

Possible causes: Jewellery is bought occasionally, and there is no sign of a follow-up process. The loyalty programme is barely used (254 payments). Selling old gold gives customers a reason to return, which may explain the higher repeat rate of buy-back customers.

> **Recommendation:** Because repeat buyers return at about one month, send a reminder or offer at **day 25–30 after a first purchase**. Learn from Branches 02 and 12 how they retain customers, and start with Branches 16 and 13. Relaunch the loyalty programme with a clear benefit for the second purchase.

5. C-03 Returns

Business question: *How often are items returned, and which branches have the most returns?*

Source queries: return_analysis

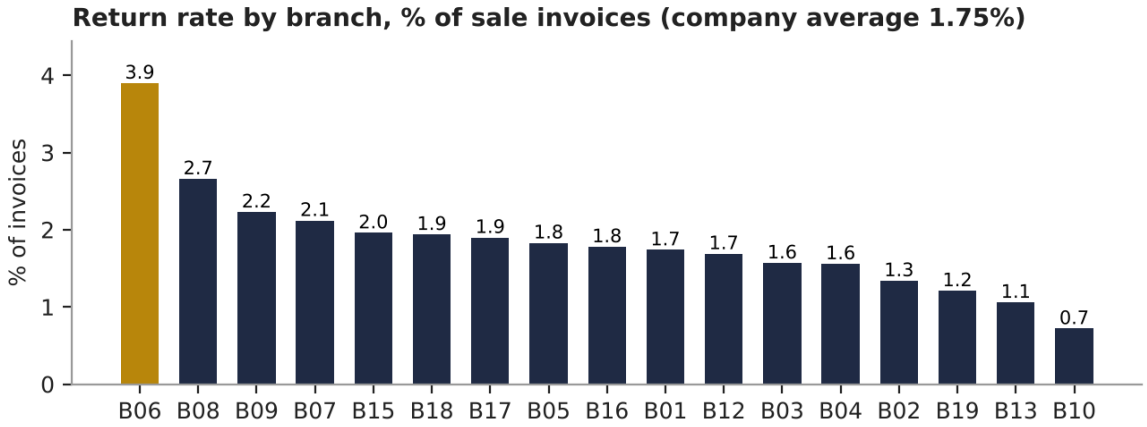


Figure 3. Returns as a percentage of sale invoices, by branch.

- **Returns are low overall:** 1,071 returns against 61,361 invoices, or **1.75%**.
- **Branch 06 is highest at 3.9%** (25 returns), more than twice the company average. It is a small branch with data for only three months, so the rate rests on few cases.
- **Branch 08 has the most returns by count (197)** at a rate of 2.7%, which matters more in practice because of its size. Branches 09 and 07 follow at 2.2% and 2.1%.
- **Branch 10 has the lowest rate (0.7%)**, followed by Branch 13 (1.1%) and Branch 19 (1.2%).

Possible causes: Small branches have few cases, so their rates swing. Branch 08 has many low-value sales, which may bring more exchanges and returns. The reason for each return is not recorded, so none of this can be confirmed.

> **Recommendation:** Focus review on Branch 08 and Branch 09, where the volume is meaningful. Start recording a **return reason** at the point of return, because the data cannot say why items come back. Check Branch 10's sales and return process as a possible model.

6. C-04 Payment Modes

Business question: *How do customers pay, and how does this affect cash handling?*

Source queries: payment_mode_split

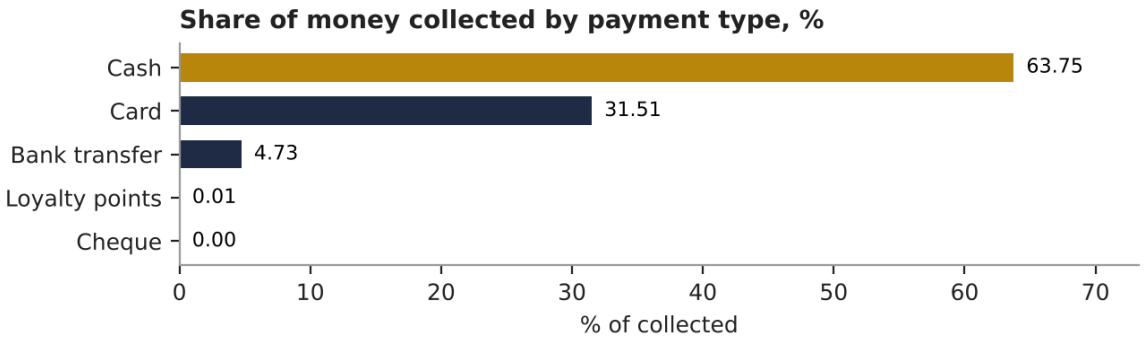


Figure 4. Share of money collected by payment type.

- **Cash is 63.8%** of money collected (SAR 224.8M from 45,103 payments).
- **Card is 31.5%** (SAR 111.1M from 26,692 payments).
- **Bank transfer is 4.7%** (SAR 16.7M) from only 1,140 payments, so transfers are used for large amounts.
- **Loyalty points and cheques are negligible** (254 and 1 payments).

Total collected (SAR 352.6M) is slightly higher than sales revenue (SAR 348.3M). This is most likely because payment records include amounts tied to return lines, and it should be confirmed.

Possible causes: Likely customer habit and the high value of gold purchases. Card acceptance by branch was not analysed.

> **Recommendation:** Cash carries handling, security and banking costs. Encourage card payment through small incentives, and check that high-value cash sales follow the company's cash-handling and anti-money-laundering procedures.

7. C-05 Old Gold Buy-Back

Business question: *Does buying back old gold lead to higher spending, and where is it used most?*

Source queries: buyback_vs_others, old_gold_activity

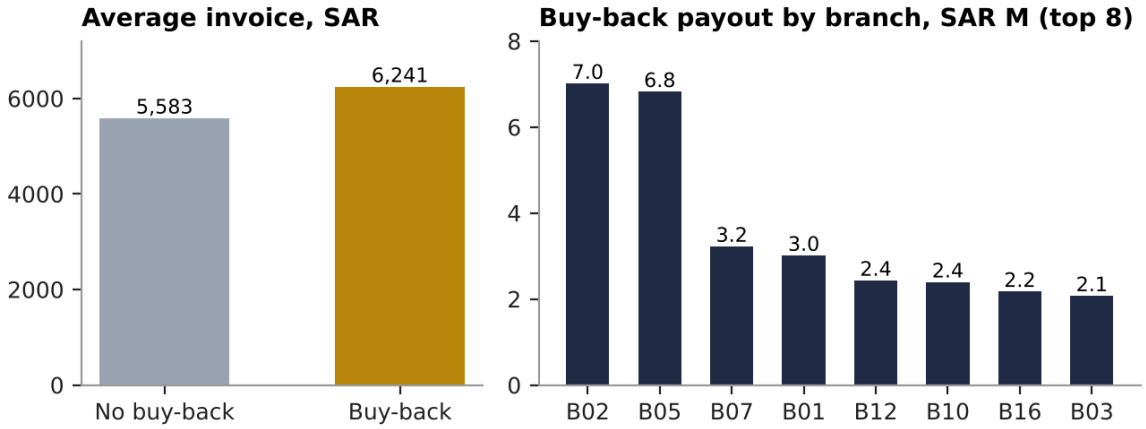


Figure 5. Average invoice value by customer group (left) and buy-back payout by branch (right).

- **Buy-back customers spend more per invoice:** SAR 6,241 against SAR 5,583, about **12% higher**. They account for 8,760 of 61,361 invoices (14%).
- **Buy-back customers also come back more often** (45.7% repeat rate against 14.6%, see Section 4).
- **Total payout is about SAR 37.8M**, equal to roughly 11% of revenue.
- **Use is concentrated.** Branches 02 and 05 account for about 37% of the payout (SAR 7.0M and 6.8M). Branch 05 also takes in the most weight (about 42,800 g).
- **Large branches use it little.** Branch 13 earns SAR 26.2M in revenue but paid out only SAR 1.1M (about 4%), while Branch 02 paid out SAR 7.0M on SAR 35.7M (about 20%).

This is an association, not proof of cause: customers who sell old gold may simply be higher-value customers to begin with.

Possible causes: Buy-back may give customers trade-in credit, which raises their spending power. Branches 02 and 05 may also run active buy-back counters. Customers who sell gold may simply be more loyal to begin with.

> **Recommendation:** Pilot Branch 02's buy-back practice (pricing, staff incentive, display) in Branches 13 and 19, then compare invoice values and repeat rates before and after.

8. C-06 Salesperson Effectiveness

Business question: *Who are the top-performing salespeople, and how concentrated is performance?*

Source queries: salesperson_ranking, plus the Avg Making Rate chart in Power BI

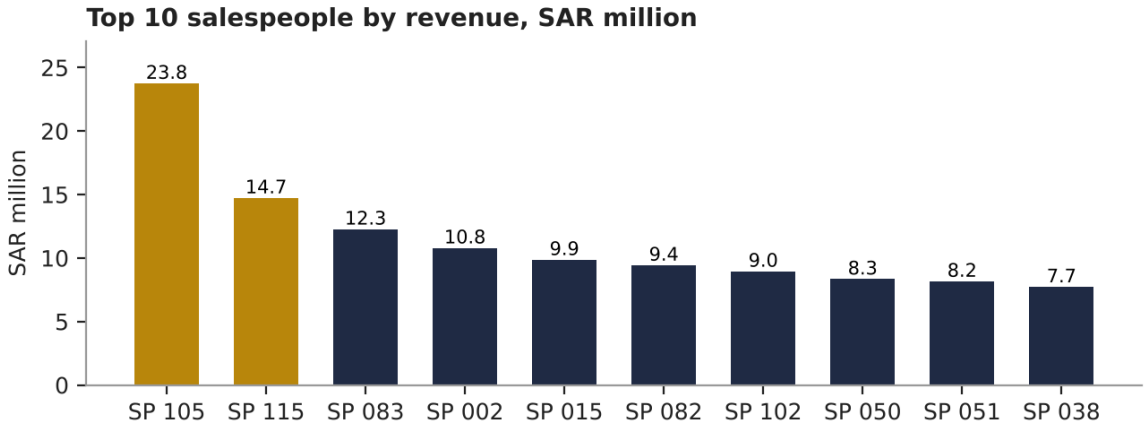


Figure 6. Top 10 salespeople by revenue. Gold bars mark IDs that equal a whole branch.

- **125 salespeople are recorded.** The top 10 generate about SAR 114M, roughly 33% of revenue.
- **Two top IDs are really branches.** Salesperson 105's revenue equals Branch 05's exactly, and Salesperson 115's equals Branch 18's. These branches appear to record every sale under one login.
- **Ticket size varies widely.** Salesperson 102 produced SAR 9.0M from only 461 invoices (about SAR 19,400 each), while Salesperson 105 averaged about SAR 5,300 across 4,483 invoices. A few people sell high-value pieces rather than high volume.
- **The bottom ten have between 1 and 11 invoices** in total, and look like transfers, trainees or test records.
- **Discounting cannot be compared.** Only 58 of about 90,800 sale lines carry a discount, so there is nothing to rank. Average making rate per gram was used instead as a measure of selling quality.

Possible causes: Branches 05 and 18 record all sales under one login, and some high-value specialists sell few but large invoices. Very low counts at the bottom suggest short tenure or test records.

> **Recommendation:** Give every salesperson an **individual login**, so rankings and incentives reflect real people. Study how Salesperson 102 sells high-value items and use it for training.

9. C-07 Daily and Seasonal Patterns

Business question: *When do sales peak, by month and by day of the week?*

Source queries: monthly_growth, monthly_trends, running_total, day_of_week

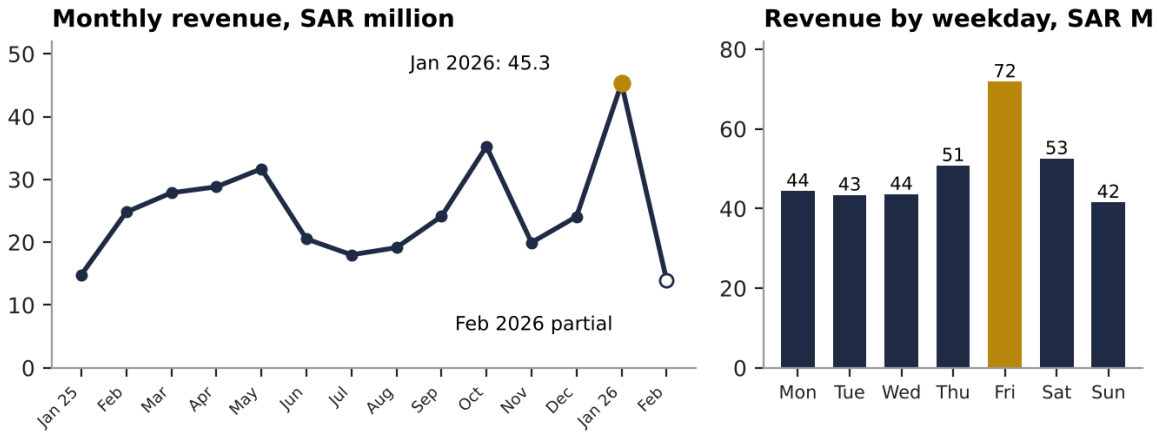


Figure 7. Monthly revenue (left) and revenue by weekday (right).

- **Peak months:** January 2026 (SAR 45.3M), October 2025 (35.3M) and May 2025 (31.7M). **Quietest:** July 2025 (18.0M); revenue stays below SAR 25M from June to September except September itself at 24.1M.
- **January 2026 was a record month for several branches** at once (Branches 03, 05, 16 and 18), which drives the +88% month-on-month growth and the −69% fall that follows in February.
- **February 2026 looks incomplete** (1,730 invoices against 4,000–5,000 in a normal month), so it should not be read as a decline.
- **Friday is the peak day:** 15,136 invoices (25% of the total) and SAR 71.9M (21%), about double a Monday, Tuesday, Wednesday or Sunday. Thursday, Friday and Saturday together bring in about 50% of revenue. Sunday is the quietest day.
- **Friday invoices are smaller:** about SAR 4,750 on average against roughly 5,800–6,300 on other days, meaning more customers buy smaller pieces.

The data does not say why months peak. Likely drivers such as holidays, campaigns and gold price movements should be checked against the calendar and the gold price history before conclusions are drawn.

Possible causes: Friday and Saturday are the weekend in Saudi Arabia, which likely brings more visitors and smaller, casual purchases. Monthly peaks could come from holidays, campaigns or gold price moves, and the data does not show which.

> **Recommendation:** Schedule the most experienced staff and the widest stock for **Thursday to Saturday**, and plan promotions ahead of October and January. Because only 14 months of data exist, year-over-year comparison is only possible for January and February.

10. C-08 Karat and Product Mix

Business question: *Which karats and item types generate revenue, and which earn the most making charge?*

Source queries: karat_mix, item_type_mix, item_type_abc_analysis

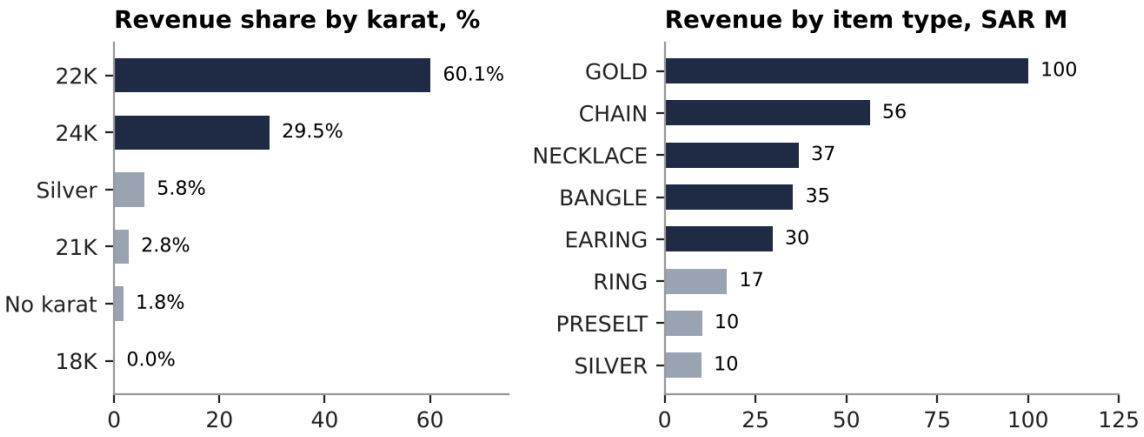


Figure 8. Revenue share by karat (left) and revenue by item type (right).

- **22K is 60.1% of revenue and 24K is 29.5%**, so these two karats make up about 90%. Silver is 5.8%, 21K 2.8% and 18K almost nothing.
- **Silver is large by weight and small by value:** it is about 71% of all weight sold but under 6% of revenue.
- **Five item types make up 75% of revenue** (Gold, Chain, Necklace, Bangle, Earring), and six with Ring reach 80%. The remaining 50-plus item types each contribute under 1%.
- **Making charge per gram differs sharply:** Ring SAR 88.7, Earring 74.6, Necklace 54.6, Chain 43.9 and Bangle 43.8, against only 11.3 for plain gold and 0.9 for silver. Diamond items show zero, because their making charge is not recorded separately.
- **Old-gold item types also appear in sales** (about SAR 14.3M, 4% of revenue). This should be confirmed with the business, since it may reflect resale of bought-back gold.

Possible causes: 22K is the usual purity for retail jewellery and 24K for bars and coins. Silver is cheap per gram, so it is large by weight but small by value. Rings and earrings are light and detailed, so the labour charge per gram is higher.

Recommendation: Protect stock availability in the top five item types, and grow the **high making-charge categories** (rings, earrings, necklaces), which earn more per gram than plain gold. Review whether the long tail of low-revenue item codes needs to be kept.

11. Summary of Recommendations

#	Recommendation	Challenge	Priority
1	Day-30 follow-up offer for first-time customers; relaunch loyalty programme; offer trade-in to one-time buyers	C-02	High
2	Pilot Branch 02's buy-back approach in Branches 13 and 19	C-05	High
3	Give every salesperson an individual login	C-06	High
4	Staff and promote for Thursday to Saturday; plan for October and January peaks	C-07	Medium
5	Start recording a return reason; review Branches 08 and 09	C-03	Medium
6	Benchmark branches on revenue per active month; confirm status of Branches 04, 06, 15, 17	C-01	Medium
7	Promote card payment and review high-value cash handling	C-04	Medium
8	Grow rings, earrings and necklaces; review the long tail of item codes	C-08	Low

12. Data Quality and Limitations

- **Short history.** Data covers January 2025 to February 2026 (the date table runs November 2024 to March 2026), and February 2026 is incomplete. Year-over-year figures only exist for January and February.
- **Partial and empty branches.** Branches 04, 06, 15 and 17 have three months of data, Branch 03 starts in July 2025, Branch 20 in January 2026, and Branches 11 and 14 have no sales.
- **Totals differ slightly between sources.** Sales including return lines (SAR 352.7M) and payments collected (SAR 352.6M) are about SAR 4.3M above sales-only revenue (SAR 348.3M), and the item-type totals sum to SAR 344.8M. This report uses the sales-only figure throughout.
- **Shared and generic identifiers.** Some salesperson IDs equal whole branches, and some customer records have hundreds of invoices, so the VIP segment, the buy-back repeat rate and some branch repeat rates may include generic walk-in records. Customer value is revenue over 14 months, not true lifetime value.
- **Discounts are almost absent** (58 of about 90,800 lines), so discount analysis is not meaningful. 258 customers have no sales, and some summary tables have gaps, so they were not used for revenue figures.
- **Missing karat.** Some sale lines have no karat and appear as "No karat" (1.8% of revenue).
- **Association, not causation.** Findings such as the buy-back uplift describe relationships in the data and should be tested before being treated as causes.

13. Conclusion

GoldVista has a strong core: a handful of branches, two karats and five product types carry most of the business, and customers who do return come back quickly. The biggest opportunities are retention (84% buy once), consistent use of old-gold buy-back across large branches, and better data capture for returns and salespeople. The Power BI dashboard and the SQL views in the project repository allow these measures to be tracked as the business acts on them.